

1D Heat Conduction

Derivation of the Heat Equation

MMAE 350

Big Picture

Goal of this module

Derive the 1D transient heat equation from

- ▶ Conservation of energy
- ▶ Fourier's law

We will also discuss:

- ▶ Boundary conditions (Dirichlet and Neumann)
- ▶ Physical meaning of parameters
- ▶ Units in SI

Physical Model

Consider a 1D bar:

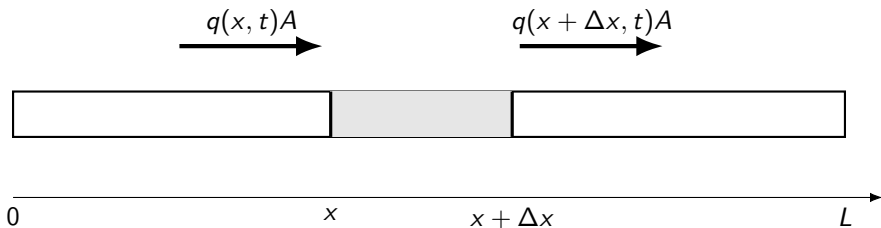
- ▶ Length L
- ▶ Constant cross-sectional area A
- ▶ Temperature $T(x, t)$

Heat flows only in the x -direction.

Assumptions

- ▶ No radiation
- ▶ Constant material properties
- ▶ Continuum model

Control Volume



- ▶ $q(x, t)$ = **heat flux** [W/m²]
- ▶ A = cross-sectional area [m²]

Energy Stored in the Control Volume

Symbol	Name	SI Units
ρ	mass density	kg/m^3
c	specific heat capacity	$\text{J}/(\text{kg}\cdot\text{K})$

Thermal energy per unit volume:

$$\rho c T \quad \left[\frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{J}}{\text{kg}\cdot\text{K}} \cdot \text{K} = \frac{\text{J}}{\text{m}^3} \right]$$

Total energy in a segment:

$$\rho c T A \Delta x \quad \left[\frac{\text{J}}{\text{m}^3} \cdot \text{m}^2 \cdot \text{m} = \text{J} \right]$$

Conservation of Energy

Rate in – Rate out = Rate of accumulation

$$q(x, t)A - q(x + \Delta x, t)A = \rho c A \Delta x \frac{\partial T}{\partial t}$$

Divide by $A\Delta x$:

$$\frac{q(x, t) - q(x + \Delta x, t)}{\Delta x} = \rho c \frac{\partial T}{\partial t}$$

Take limit as $\Delta x \rightarrow 0$:

$$-\frac{\partial q}{\partial x} = \rho c \frac{\partial T}{\partial t}$$

Fourier's Law

Constitutive law

$$q = -k \frac{\partial T}{\partial x}$$

k = thermal conductivity

SI units:

$$k \text{ [W/(m}\cdot\text{K)]}$$

Heat flows from hot to cold.

Derivation of the Heat Equation

Substitute Fourier's law:

$$-\frac{\partial}{\partial x} \left(-k \frac{\partial T}{\partial x} \right) = \rho c \frac{\partial T}{\partial t}$$

If k is constant:

$$k \frac{\partial^2 T}{\partial x^2} = \rho c \frac{\partial T}{\partial t}$$

Rearrange:

$$\boxed{\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}}$$

$$\alpha = \frac{k}{\rho c}$$

Thermal diffusivity:

$$\alpha [\text{m}^2/\text{s}]$$

Physical Meaning of Parameters

- ▶ k : how easily heat flows through a material
- ▶ c : how much energy required to change temperature
- ▶ ρ : how much material per unit volume
- ▶ α : how fast temperature disturbances spread

High diffusivity $\rightarrow$ temperature equalizes quickly.

Boundary Conditions

To solve the PDE, we need:

Dirichlet (Prescribed Temperature)

$$T(0, t) = T_0$$

Neumann (Prescribed Heat Flux)

$$-k \frac{\partial T}{\partial x} = q^*$$

Insulated Boundary

$$\frac{\partial T}{\partial x} = 0$$

Initial Condition

Since the equation is time-dependent, we also need:

$$T(x, 0) = f(x)$$

The PDE + BCs + IC completely define the problem.

Summary

- ▶ Derived 1D heat equation from energy conservation
- ▶ Introduced Fourier's law
- ▶ Defined thermal diffusivity
- ▶ Discussed boundary conditions

Next:

- ▶ Steady-state solutions
- ▶ Finite difference discretization
- ▶ FTCS method